



Quiz game



This is a C program that displays a menu-driven quiz game. Here is a header files that are used in the program. These include standard input/output functions.

there are function prototypes for four functions that are defined later in the code: `show_record()`, `reset_score()`, `help()`, and `edit_score()`.

- The `main()` function is the starting point of the program. It begins by setting the console color to a specific value using the `system()` function.
- The program declares three variables: a `float` variable called `score`, a `char` variable called `choice`, and a character array called `playername` with a size of 20.
- The program then declares some integer variables: `count`, `r`, `r1`, `count`, `i`, and `n`. These variables are used later in the program.
- The program then uses a label called `mainhome` to create a loop that allows the user to return to the main menu after completing a quiz or performing another action.
- The `system("cls")` function clears the console screen at the beginning of each iteration of the loop.

- | | |
|---|--|
| • <code>#include<stdio.h></code> | • <code>int main()</code> |
| • <code>#include<conio.h></code> | • <code>{</code> |
| • <code>#include<ctype.h></code> | • <code>system ("COLOR 56");</code> |
| • <code>#include<stdlib.h></code> | • <code>float score;</code> |
| • <code>#include<string.h></code> | • <code>char choice;</code> |
| • <code>void show_record();</code> | • <code>char playername[20];</code> |
| • <code>void reset_score();</code> | • <code>int count,r,r1,count,i,n;</code> |
| • <code>void help();</code> | • <code>mainhome:</code> |
| • <code>void edit_score(float score, char choice);</code> | • <code>system("cls");</code> |

- If the user chooses 'V', the `show\_record()` function is called, which displays the high scores saved in a file. Then, the program jumps to the label `mainhome` using the `goto` statement.
- If the user chooses 'H', the `help()` function is called, which provides instructions on how to play the game. Then, the program jumps to the label `mainhome` using the `goto` statement.
- If the user chooses 'R', the `reset\_score()` function is called, which clears the high scores saved in a file. Then, the program jumps to the label `mainhome` using the `goto` statement.
- If the user chooses 'Q', the program exits using the `exit(1)` statement.
- If the user chooses 'S', the user interface screen is cleared using the `system("cls")` statement. Then, the `edit\_score()` function is called, which allows the user to enter their name and score and save it to a file.

- `choice=toupper(getch());`
- `system("cls");`
- `printf("\t\t\tC PROGRAM QUIZ GAME\n");`
- `printf("\n\t\t_____`
`_____");`
- `printf("\n\t\t\t WELCOME ");`
- `printf("\n\t\t\t to ");`
- `printf("\n\t\t\t THE GAME ");`
- `printf("\n\t\t_____`
`_____");`
- `printf("\n\t\t_____`
`_____");`
- `printf("\n\t\t BECOME A MILLIONAIRE!!!!!!!!!!!! ");`
-

- `printf("\n\t\t_____`
`_____");`
- `printf("\n\t\t > Press S to start the game");`
- `printf("\n\t\t > Press V to view the highest score ");`
- `printf("\n\t\t > Press R to reset score");`
- `printf("\n\t\t > press H for help ");`
- `printf("\n\t\t > press Q to quit ");`
- `printf("\n\t\t_____`
`_____ \n\n");`
- `choice=toupper(getch());` |
- `if (choice=='V')`
- `{`
- `show_record();`
- `goto mainhome;`
- `}`
- `else if (choice=='H')`

- `if (toupper(getch())=='Y')`
- `{`
- `goto home;`
- `}`
- `else`
- `{`
- `goto mainhome;`
- `system("cls");`
- `}`
- `home:`
- `system("cls");`
- `count=0;`
- `for(i=1;i<=3;i++)`
- `{`
- `system("cls");`

The program asks the user if they want to play the quiz again. If the user presses 'Y' (in uppercase), the program jumps to the `home` label using the `goto` statement. Otherwise, the program jumps to the `mainhome` label, clears the user interface screen using the `system("cls")` function, and continues executing the program.

- The `toupper()` function is used to convert the user's input to uppercase, allowing the program to detect whether the user entered 'Y' or 'y'.

- The `getch()` function is used to read a single character of input from the keyboard without echoing it to the screen.

- The `system("cls")` function is used to clear the user interface screen.

- `r1=i;`
- `switch(r1)`
- `{`
- `case 1:`
- `printf("\n\nWhich of the following is a Palindrome number?");`
- `printf("\n\n1.42042\t\t2.101010\n\n3.23232\t\t4.01234");`
- `if (toupper(getch())=='3')`
- `{`
- `printf("\n\nCorrect!!!");count++;`
- `getch();`
- `break;`
- `}`
- `else`
- `{`
- `printf("\n\nWrong!!! The correct answer is C.23232");`

- The ``switch`` statement is used to determine which question to ask the user. Several functions and statements are used to display the question and options, and verify the user's answer.
- - The variable ``r1`` is used to store the number of the question to be asked.
- - The ``printf`` statements are used to display the question and options to the user.
- - The ``getch`` function is used to read the user's input from the keyboard.
- - The ``toupper`` function is used to convert the user's input to uppercase, allowing the program to detect whether the user entered the correct answer.
- - The ``printf`` statements are used to display the message "Correct!!!" if the user enters the correct answer, and "Wrong!!!" if the user enters an incorrect answer.
- - The variable ``count`` is used to keep track of the number of correct answers entered by the user.
- - The ``break`` statement is used to exit the current case and move to the next case in the ``switch`` statement.

- `if(count>=2)`
- `{goto test;}`
- `else`
- `{`
- `system("cls");`
- `printf("\n\nSORRY YOU ARE NOT ELIGIBLE TO PLAY THIS GAME, BETTER LUCK NEXT TIME");`
- `getch();`
- `goto mainhome;`
- `}`
- `test:`
- `system("cls");`
- `printf("\n\n\t*** CONGRATULATION %s you are eligible to play the Game ***",`
- `playername);`
- `printf("\n\n\n\n\t!Press any key to Start the Game!");`
- `printf("\n\n\n\n\t!Press any key to Start the Game!");`
- `if(toupper(getch())=='p')`
- `{goto game;}`
- `game:`
- `countr=0;`
- `for(i=1;i<=10;i++)`
- `{system("cls");`

In this part of the code, the program checks whether the user has answered at least two questions correctly. If the user has answered at least two questions correctly, the program jumps to the `test` label. Otherwise, the program displays a message informing the user that they are not eligible to play the game and returns to the main menu using the `goto` statement.

- The `count` variable is used to keep track of the number of correct answers entered by the user.

- The `goto` statement is used to jump to the `test` label if the user has answered at least two questions correctly, or to return to the main menu using the `goto` statement if the user has answered less than two questions correctly.

- The `printf` and `system` functions are used to display appropriate messages and clear the user interface screen.

- The `getch` function is used to read the user's input from the keyboard.

- The `toupper` function is used to convert the user's input to uppercase, allowing the program to detect whether the user entered 'p' to start the game.

- The `game` label is used to start the game.

- The `countr` variable is used to keep track of the number of correct answers entered by the user in the game.

- The loop `for` is used to present the user with a series of ten questions.

- for(i=1;i<=10;i++)
- {system("cls");
- r=i;
- switch(r)
- {
- case 1:
- printf("\n\nWhat is the National Game of England?");
- printf("\n\nA.Football\t\tB.Basketball\n\nC.Cricket\t\tD.Baseball");
- if (toupper(getch())=='C')
- {printf("\n\nCorrect!!!");countr++;getch();
- break;getch();}
- else
- {printf("\n\nWrong!!! The correct answer is C.Cricket");getch();
- goto score;
- case 2:
- break;}
- printf("\n\nStudy of Earthquake is called.....,");
-
- printf("\n\nA.Seismology\t\tB.Cosmology\n\nC.Orology\t\tD.Etimology");
- if (toupper(getch())=='A')
- {printf("\n\nCorrect!!!");countr++;getch();
- break;}
- else
- {printf("\n\nWrong!!! The correct answer is A.Seismology");getch();
- goto score;

A "for" loop is used to display a series of ten questions to the user. A "switch" statement is used to determine which question to ask the user. Several statements and functions are used to display the question and choices, and to check the user's answer.

- The variable "i" is used in the "for" loop to keep track of the number of questions asked to the user.

- The "system" statement is used to clear the user interface screen each time a new question is asked.

- The variable "r" is used in the "switch" statement to determine which question to ask the user.

- The "printf" statements are used to display the question and choices to the user.

- The "getch" statement is used to read the user's input from the keyboard.

- The "toupper" function is used to convert the user's input to uppercase letters.

- The "if" statement is used to check the user's answer.

- The variable "countr" is used to keep track of the number of correct answers entered by the user.

- The "printf" statement is used to display the message "Correct!!!" if the user enters a correct answer, and to display the message "Wrong!!!" if the user enters an incorrect answer.

- The "goto" statement is used to jump to the "score" label if the user enters a incorrect answer.

- score:
- system("cls");
- score=(float)count*100000;
- if(score>0.00 && score<1000000)
- {
- printf("\n\n\t\t***** CONGRATULATION
*****");
- printf("\n\t You won \$%.2f",score);goto go;}
- else if(score==1000000.00)
- {
- printf("\n\n\n\t\t***** CONGRATULATION
*****");
- printf("\n\t\t\t\t YOU ARE A MILLIONAIRE!!!!!!!!!!");
- printf("\n\t\t You won \$%.2f",score);
- printf("\n\n\t\t***** SORRY YOU DIDN'T WIN ANY CASH
*****");
- }
- else
- {
- go:
- puts("\n\n Press Y if you want to play next game");
- puts(" Press any key if you want to go main menu");
- if (toupper(getch())=='Y')
- goto home;
- else
- {

After calculating the user's score, the "score" label is used to display a congratulatory message to the user and give them the option to play again or return to the main menu.

- The "system" function is used to clear the user interface screen.
- The "score" variable is calculated based on the number of correct answers the user has entered.
- An "if" statement is used to display a congratulatory message and the user's score if the score is greater than zero and less than one million dollars.
- An "else if" statement is used to display a special congratulatory message and the user's score if the score is equal to one million dollars.
- An "else" statement is used to display a message indicating that the user did not win any cash.
- The "goto" statement is used to move to the "go" label.
- The "puts" function is used to display an encouraging message to the user.
- An "if" statement is used to move to the "home" label if the user wants to play again.
- An "else" statement is used to move to the "mainhome" label if the user wants to return to the main menu.
- The "edit_score" function is used to update the user's score in the database.

- `void show_record()`
- `{system("cls");`
- `char name[20];`
- `float scr;`
- `FILE *f;`
- `f=fopen("score.txt","r");`
- `fscanf(f,"%s%f",&name,&scr);`
- `printf("\n\n\t\t*****`
`*****");`
- `printf("\n\n\t\t%s has secured the Highest Score`
`%0.2f",name,scr);`
- `printf("\n\n\t\t*****`
`*****");`
- `fclose(f);`
- `getch();}`

- The function "show_record" is used to display the highest score achieved by any player. The function reads the "score.txt" file and retrieves the name and score of the player with the highest score. The retrieved values are then displayed on the screen.
- - The "system" function is used to clear the user interface screen.
- - The "name" and "scr" variables are declared to store the name and score of the player with the highest score.
- - The "FILE" pointer "f" is declared and initialized to point to the "score.txt" file.
- - The "fscanf" function is used to read the name and score of the player with the highest score from the "score.txt" file.
- - The retrieved values are displayed on the screen using the "printf" function.
- - The "fclose" function is used to close the "score.txt" file.
- - The "getch" function is used to wait for the user to press a key before exiting the function.

- `void reset_score()`
- `{system("cls");`
- `float sc;`
- `char nm[20];`
- `FILE *f;`
- `f=fopen("score.txt","r+");`
- `fscanf(f,"%s%f",&nm,&sc);`
- `sc=0;`
- `fprintf(f,"%s,%.2f",nm,sc);`
- `fclose(f);}`

- The function "reset_score" is used to reset the highest score to zero. The function reads the "score.txt" file, sets the score to zero, and then writes the new score to the file.
- - The "system" function is used to clear the user interface screen.
- - The "sc" and "nm" variables are declared to store the score and name of the player with the highest score.
- - The "FILE" pointer "f" is declared and initialized to point to the "score.txt" file.
- - The "fscanf" function is used to read the name and score of the player with the highest score from the "score.txt" file.
- - The "sc" variable is set to zero. This is done to reset the highest score to zero. After that, the new name and score (which is zero) are written to the "score.txt" file using the "fprintf" function, and the file is closed using the "fclose" function. Once the game is played again, the user will have a new opportunity to achieve a higher score than zero and update it in the "score.txt" file.
- - The "fprintf" function is used to write the new name and score to the "score.txt" file.

- void help()
- {system("cls");
- printf("\n\n\t\t\t\t\tHELP");
- printf("\n -----"
-);
- printf("\n C program Quiz Game.....");
- printf("\n >> There are two rounds in the game, WARMUP ROUND & CHALLENGE ROUND");
- printf("\n >> In warmup round you will be asked a total of 3 questions to test your general");
- printf("\n\t knowledge. You will be eligible to play the game if you can give
- atleast 2");
- printf("\n\t right answers otherwise you can't play the Game.....");
- printf("\n >> Your game starts with the CHALLENGE ROUND. In this round you will be
- asked");
- printf("\n\t total 10 questions each right answer will be awarded \$100,000.");
- printf("\n\t By this way you can win upto ONE MILLION cash prize in
- USD.....");
- printf("\n >> You will be given 4 options and you have to press A, B ,C or D for the"
-);
- printf("\n\t right option");
- printf("\n >> You will be asked questions continuously if you keep giving the right answers.");

- The "help" function is used to display instructions and guidelines for playing the quiz game.
- - The "system" function is used to clear the user interface screen.
- - The "printf" function is used to display the instructions on the screen.
- - The instructions include information about the two rounds of the game, the warmup round and the challenge round.
- - In the warmup round, the player is asked 3 questions to test their general knowledge. They must answer at least 2 questions correctly to be eligible to play the game.
- - The challenge round consists of 10 questions, and each correct answer earns the player \$100,000. The player can win up to \$1,000,000 in total.
- - The player is given 4 options to choose from and they must press A, B, C, or D for the correct option.
- - The player will be asked questions continuously as long as they keep giving the right answers.
- - There is no negative marking for wrong answers.
- - The function ends by wishing the player good luck and acknowledging that the C Program Quiz Game was developed by the Code With C team.

- `void edit_score(float score, char plnm[20])`
- `{system("cls");`
- `float sc;`
- `char nm[20];`
- `FILE *f;`
- `f=fopen("score.txt","r");`
- `fscanf(f,"%s%f",&nm,&sc);`
- `if (score>=sc)`
- `{ sc=score;`
- `fclose(f);`
- `f=fopen("score.txt","w");`
- `fprintf(f,"%s\n%.2f",plnm,sc);`
- `fclose(f);}}`

- The "edit_score" function is used to update the highest score in the "score.txt" file if the current score is higher than the score in the file.
- - The "system" function is used to clear the user interface screen.
- - A variable "sc" is created to store the current score from the file.
- - A variable "nm" is created to store the name of the player who achieved the highest score in the file.
- - The "score.txt" file is opened using the "fopen" function with "r" mode (for reading).
- - The "fscanf" function is used to read the name and the current score from the file.
- - If the current score (which is passed as an argument to the function) is higher than the score in the file, then the new score replaces the old score in the file.
- - The file is closed using the "fclose" function.
- - The "score.txt" file is opened again using the "fopen" function with "w" mode (for writing).
- - The "fprintf" function is used to rewrite the name and the new score to the file.
- - The file is closed again using the "fclose" function.